Effectiveness of topical skin products in the treatment and prevention of incontinence-associated dermatitis: a systematic review

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EXECUTIVE SUMMARY

Background
Incontinence-associated dermatitis (IAD) is inflammation of the skin resulting from repeated contact with urine and/or feces. It causes pain, redness, swelling and excoriation, and may lead to complications such as fungal skin infections and pressure injuries. It is important to prevent and treat IAD to avoid complications and improve patient outcomes. A number of products are available for protecting skin, but evidence on their effectiveness is limited.

Objectives
The current review aimed to establish the effectiveness of topical skin products in reducing the occurrence and/or severity of IAD.

Inclusion criteria
Types of participants
Adult patients over the age of 18 years with incontinence and/or diarrhea.

Types of intervention
Topical skin products as individual interventions or part of a skin care regimen.

Types of studies
Both published and unpublished study designs including randomized controlled trials, non-randomized controlled trials, quasi-experimental, before and after, prospective and retrospective cohort, case-control, analytical cross-sectional, descriptive study designs including case series, individual case reports and descriptive cross-sectional studies across all care settings for inclusion.

Outcomes
The primary outcome of interest was the absence or non-development, reduction or resolution, new development or increase in the occurrence of IAD or the increase in severity of IAD. The secondary outcome was any adverse effect caused by the skin care products used.

Search strategy
A three-step search strategy to find both published and unpublished papers was utilized in this review. Studies were limited to those published in English from 1980 to 2016.

Methodological quality
Papers selected were assessed by two independent reviewers using the Joanna Briggs Institute Meta-Analysis of Statistics Assessment and Review Instrument (JBI-MAStARI).

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Data extraction
Data were extracted using the standardized data extraction tool in JBI-MAStARI. The data extracted included specific details about the interventions, populations, study methods and outcomes.

Data synthesis
Studies were assessed for methodological quality and statistical significance to determine validity and generalizability of study results. It was not possible to pool studies to conduct meta-analysis or test for heterogeneity.

Results
There were a limited number of clinical trials that compared products for efficacy in preventing and treating IAD. Assessment tools and severity scores used to measure skin damage outcomes were dissimilar and prone to subjectivity. It was difficult to ascertain superiority of any individual product.

Conclusion
Information on barrier protective efficacy, side effects and cost can be valuable to both clinicians and care providers. More randomized controlled trials on product effectiveness for prevention and treatment of IAD are highly recommended.

Keywords Incontinence-associated dermatitis; nappy/napkin rash; perineal dermatitis; rash; treatment

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Background

Incontinence-associated dermatitis (IAD) is an inflammation of the skin that occurs when the skin comes into prolonged contact with urine and/or fecal matter.[1] It is categorized as an inflammatory skin condition that presents with redness, swelling, excoriation and pain, and may be complicated by opportunistic fungal infections.[2] The inflammation is usually confined to skin areas that come into contact with urine and/or feces and can include the perianal and gluteal cleft, the buttocks, upper thighs, external genitalia and suprapubic areas.[3] It differs from pressure injuries in that it is "top down" damage where superficial damage to the epidermis occurs.[4] Pressure injury on the other hand is a "bottom up" injury where deeper tissues become affected first by decreased blood flow and then progresses to the superficial tissues and the skin.[4] Incontinence and the problems associated with it in both pediatric and adult populations have been an area requiring significant resources in terms of nursing hours and costs, and continues to be a daily challenge for healthcare professionals in all settings.[5] Among the numerous difficulties associated with incontinence, skin complications are a priority because of the pain, skin infections, sepsis and pressure injuries that can result.[2,6-12]

The natural acidic pH of the skin ranges between 4 and 6.8 with a mean of 5.5. This pH range is referred to the as the "protective acid mantle".[13] Prolonged exposure to moisture, urine or feces (particularly, liquid fecal matter) causes the skin pH to become more alkaline.[12] This alters the skin's acid mantle, making it more permeable to irritants and vulnerable to bacteria and fungi.[12] Skin with an impaired barrier function and abnormal transepidermal water loss becomes more permeable, erythematous, taut and painful, and is easily abraded and excoriated in the presence of mechanical forces, for example, shearing associated with cleansing or friction.[11] This results in a red or erythematous rash with or without excoriation of skin that can be classified as mild, moderate or severe IAD.[14]

It is difficult to gauge the true scale of the problem as IAD data on both prevalence and incidence vary widely.[1,20] In instances where IAD data are reported, the prevalence is estimated at 5.6–50% and incidence at 3.4–25%.[1,6,15,20,21] There is often confusion between these two measures with inaccurate reporting and a lack of coordination of epidemiological studies among all care settings, both local and global. Most researchers focusing on IAD recommend further studies with solid research designs and IAD-specific diagnostic tools to attain more reliable data.[2,5,16-18]

The causative factor for IAD is incontinence where IAD risk is increased with the type of incontinence (fecal incontinence, urinary incontinence or double incontinence [fecal and urinary] and the

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frequency of episodes of incontinence [especially fecal] and diarrhea). Risk factors for IAD development include the use of occlusive containment products, compromised skin (e.g. due to aging, steroid use or diabetes), immobility, diminished cognitive awareness, inability to perform personal hygiene, raised body temperature (pyrexia), medications (antibiotics and immunosuppressants), poor nutritional status and critical illness.[6,19,20]

Prevention and treatment of IAD is focused on effective continence clean-up or removing bodily fluids, minimizing the length of exposure to fecal matter, redirecting the fecal matter away from the skin and applying skin protection or some form of barrier.[11] Studies have been carried out using a number of products that range from petrolatum, zinc oxide, acrylate polymer and dimethicone.[11]

Currently, practices in some clinical settings appear to be based on availability of products, and there is confusion and a lack of evidence that supports the efficacy of one product over another.[5,21] As new products emerge on the market, it can become more difficult to make choices as cost, marketing and commercialization may drive selection of one product over another.[22] Despite increasing recognition and efforts to manage the dermatological issues and complications of incontinence, to date there is a lack of rigorously performed research addressing the effectiveness of different skin care products to prevent and treat IAD.[5,21]

The identification of a product or type of product that best mimics the skin's barrier function, with a pH close to that of intact healthy skin and that remains on the skin as a waterproof protectant, is fundamental to reducing the development of IAD.[23] Products vary in formula and application method.[24] They may be packaged as a liquid, emulsion, cream, spray, foam, gel, towelette, wipe or washcloth.[10] Products that remain on the skin, as a waterproof barrier, isolate and protect the skin from enzymes in urine and feces and that do not rub off onto bed linen and clothes appear to be most effective.[12]

Limited literature reviews on comparison of effectiveness of products were retrieved during initial searching. Two reviews that focused on IAD treatment and prevention products were retrieved.[22,24] These studies are literature reviews, and although they provide insight into studies carried out on IAD treatment and prevention, they lack the robust, comprehensive, systematic analysis and testing for heterogeneity used by a systematic review design. Cost analysis studies on different products have also been carried out, and although they provide some evaluation on products and are valuable for financial comparisons, they do not inform comprehensively on the efficacy of the products.

These studies have recognized that much more research and experimental studies are needed to identify the products that have the best results in preventing or treating IAD, making a systematic review of current literature a logical starting point. An initial search of the Cochrane Library, the *JBI Database of Systematic Reviews and Implementation Reports* and MEDLINE identified no other systematic reviews on this topic.

The methods of this review were pre-specified according to *a priori* published protocol.[25]

Objectives

The current review aimed to answer the following question: is the use of topical skin products in patients with incontinence effective in reducing the occurrence of IAD or the risk of developing IAD in adult patients with fecal and/or urinary incontinence and/or diarrhea and if so which products are the most effective?

The quantitative objective for this review was to identify the effectiveness of the use of topical skin products on: i) preventing and reducing the risk of developing IAD; and ii) reducing the occurrence and severity of IAD in adult patients experiencing incontinence and/or diarrhea.

More specifically, the objective was to identify the efficacy of individual topical skin products on the IAD incident rate in adult patients who are incontinent or have diarrhea by comparing individual skin care products with either other skin care products or with a control group.

Inclusion criteria

Types of participants

The current review included studies conducted across all settings (acute care, intensive care, aged care, palliative care and community care) that included adult patients (both male and female) over the age of 18 years with urine incontinence, fecal incontinence with or without diarrhea, both urine and fecal incontinence with or without diarrhea, and diarrhea by any causal factor. For the purpose of this study, incontinence was defined as "the involuntary

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or inappropriate passing of urine and/or feces that has an impact on social functioning or hygiene".[26(p.7)] Diarrhea was defined using the Bristol Stool Form Scale.[27] Both incontinence and diarrhea were acknowledged to be present as identified by the researchers in the articles reviewed.

As there are no standardized diagnostic tools currently used to diagnose IAD, patients who had developed redness, inflammation, swelling, blistering or excoriation of skin in contact with urine and/or feces (as identified by the researchers in literature reviewed) were considered as having IAD. This included the skin in the perineal, buttocks, genital, suprapubic and upper thigh regions.

The review included studies on the use of topical skin care products or barrier creams to prevent IAD in patients who did not have IAD but were considered at risk of developing IAD (experiencing incontinence and/or diarrhea) and skin care products used to treat IAD in patients who were diagnosed with or considered to have IAD as per the criteria stipulated.

The review excluded studies on individuals under the age of 18 years, in line with the definition of "adult" and in view of the pathophysiology and etiology of pediatric or infant nappy rash being different to adult IAD.[2]

Types of interventions

The current review considered clinical studies that used topical skin products for IAD and considered which was most effective based on the following:

- The application method of skin products in liquid, emulsion, cream, spray, foam, gel, towelette, wipe or washcloth forms, and compared the effectiveness of each application method to other application methods.
- The active ingredients in the commonly used skin products, such as petrolatum, zinc oxide, acrylate polymer and dimethicone, were evaluated, and the effectiveness of each product was compared to the effectiveness of active ingredients in other products.
- The frequency of the applications, for example, once daily or "application after each incontinent episode", was reviewed to establish which frequency method was most effective.
- Care bundles or skin care regimens with or without the use of a skin care product were analyzed for effectiveness and compared to determine more or less effectiveness.

Types of studies

The review considered both published and unpublished experimental study designs including randomized controlled trials, non-randomized controlled trials, quasi-experimental studies, before and after studies, prospective and retrospective cohort studies, case-control studies and analytical cross-sectional studies for inclusion. The review also considered descriptive study designs including case series, individual case reports and descriptive cross-sectional studies for inclusion.

Outcomes

The current review considered studies that included the following primary outcome measures associated with skin care products used to treat or prevent IAD:

- The absence or non-development of IAD (redness, inflammation, swelling, blistering or excoriation of skin in contact with urine and/or feces in the perineal, buttock, genital, suprapubic and upper thigh regions).
- A reduction or the resolution of IAD.
- New development of or increase in the occurrence of IAD.
- An increase in the severity of IAD symptoms.
- The number of days from commencement of application of skin protectants to resolution of IAD symptoms for products used that demonstrated a restorative effect on skin diagnosed with IAD.

Secondary outcome measures of interest were:

- Adverse effects caused by or associated with topical skin care products used to treat or prevent the development of IAD.

Search strategy

The search strategy aimed to find both published and unpublished studies. A three-step search strategy was utilized in this review. An initial limited search of MEDLINE (via EBSCOHost) was undertaken followed by an analysis of the text words contained in the title and abstract, and of the index terms used to describe the article. A second search using all identified keywords and index terms using Boolean Operators was then undertaken on February 19, 2015 by one reviewer (Appendix I) and updated

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in 2016, applying the same search strategy. Third, the reference lists of all identified reports and articles were searched for additional studies. Studies published in English were considered for inclusion in this review, and studies in languages other than English were excluded as there were no resources available for translation. Studies published from 1980 to 2016 were considered for inclusion in this review as this appears to be the period when initial literature on the subject of IAD was published.

The databases searched included CINAHL, Cochrane Central Register of Controlled Trials, Embase, EBSCOHost, MEDLINE (via EBSCOHost), ProQuest Health and Medical Complete and Web of Science. The search for unpublished studies included Open Gray, Science.gov, Thesis and Dissertations, Clinical Trials.gov and International Standard Randomized Controlled Trial Number Registry.

Initial keywords used were: incontinence-associated dermatitis, perineal dermatitis, incontinence dermatitis, perineal rash, nappy/napkin rash and treatment. Following completion of the searches, titles and abstracts were screened by two reviewers independently and articles for retrieval identified. Retrieved articles that met the inclusion criteria were then subject to the critical appraisal process.

The full search strategy is provided in Appendix I.

Method of the review

Quantitative papers selected for retrieval were assessed by two independent reviewers for methodological validity prior to inclusion in the review using standardized critical appraisal instruments from the Joanna Briggs Institute Meta-Analysis of Statistics Assessment and Review Instrument (JBI-MAStARI) (Appendix II). Any disagreements that arose between the reviewers were resolved through discussion; a third reviewer, although available, was not required to resolve disparities.

Data extraction

Quantitative data were extracted from papers included in the review by two reviewers using the standardized data extraction tool from JBI-MAStARI (Appendix III). The data extracted included specific details about the interventions, populations, study methods and outcomes of significance to the review question and specific objectives.

Data synthesis

Statistical pooling was not possible. Heterogeneity between the studies was very high because comparators in the studies were varied and outcomes that measured skin damage severity scores were different and prone to subjectivity. In some instances, pressure injury scoring tools were used, making the findings unreliable as redness due to pressure and IAD differed in their location, appearance, pathophysiology and cause. The findings have thus been presented in narrative form including tables and figures to aid in data presentation where appropriate.

Results

Description of studies

The searches resulted in 2193 articles being retrieved through systematic database searches and 13 studies selected from reference list searches of the included articles. After duplicates were removed, the titles of the remaining 1658 articles were screened and following exclusion by title and abstract, 228 remained. These were filtered for animal studies, magazine articles and pediatric studies, and 53 articles remained for full-text screening. Study verification for suitability was carried out by two reviewers on these 53 articles. Twenty studies met the criteria and were critically appraised. Following critical appraisal, 10 studies were excluded (Appendix IV). Reasons for exclusion were outside inclusion date, not population of interest, methodology not research, laboratory studies with healthy volunteers using synthetic urine on forearms, study protocols with no results, not in English, not intervention or condition being reviewed and unclear measurement of IAD. Ten studies met the criteria for inclusion and were included in the review[7,15,19,28-34] (see Appendix V and Figure 1).

Of the 10 studies included in this review, five focused on both cost effectiveness and clinical effectiveness with the others focusing solely on clinical benefits. A total of 804 participants across all studies were included in the final review. Of these included studies, three were randomized controlled trials,[15,28,31] five quasi-experimental studies,[7,19,29,30,34] one prospective cohort descriptive study[32] and one prospective descriptive two-group study.[33] One study was unpublished but presented as a poster.[29] Of these, one study compared the effectiveness of zinc oxide oil (ZnO) against 3M™ Cavilon No Sting Barrier Film™ (NSBF)

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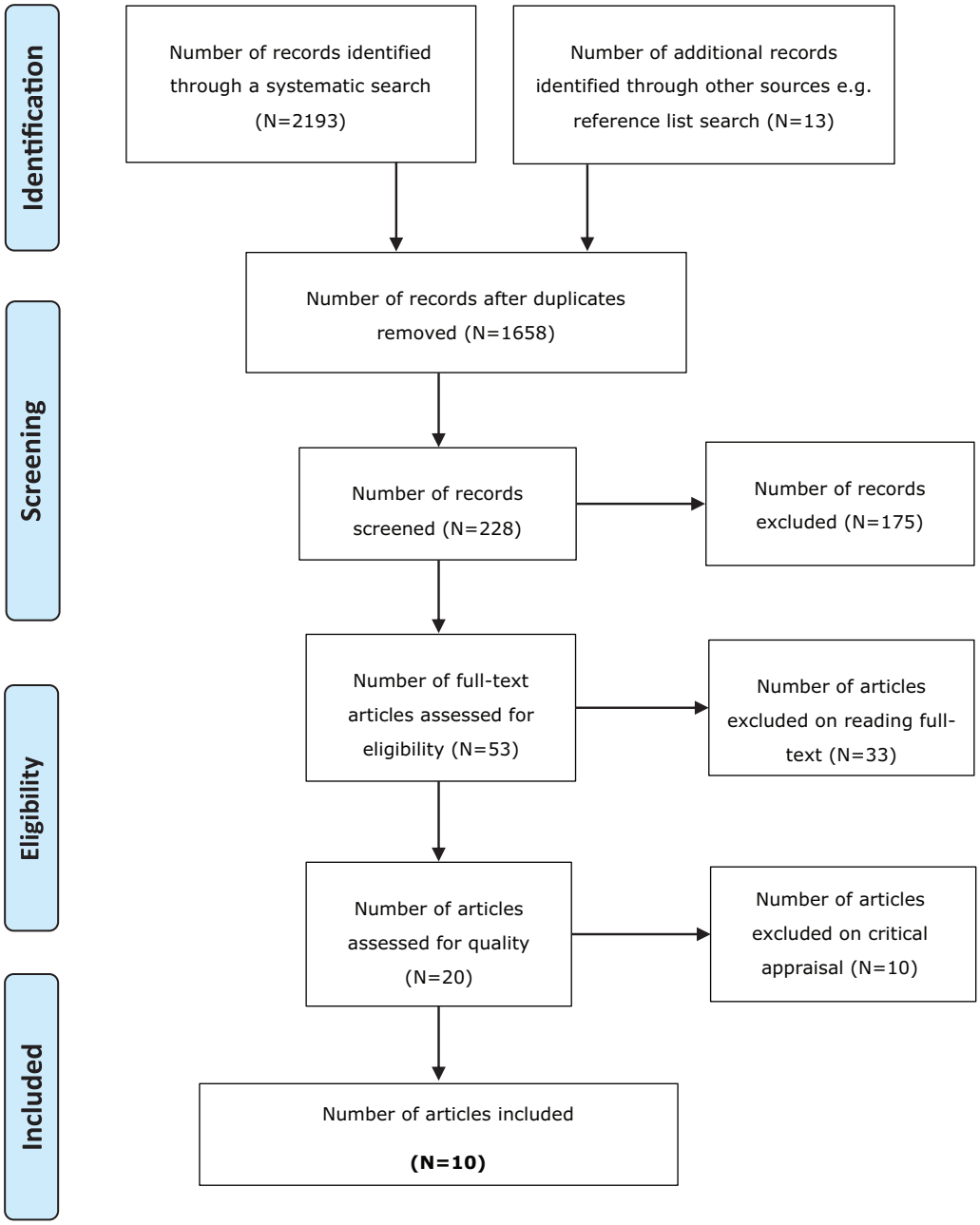

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097.

Figure 1: Literature search and study selection flow diagram

(3M Australia, North Ryde, NSW); four studies compared the effectiveness of Sage Comfort Shield® 3-in-1 wash cloths (Independence Australia Group, Independence, West Footscray, Victoria, Australia) against pH neutral soap and water, 3M™ Cavilon NSBF, no rinse cleanser and Calmoseptine® (Calmoseptine, Inc., Huntington Beach, CA) barrier cream and a standard care regimen, respectively. The study with the standard care regimen did not provide details of the care regimen. Another study

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investigated the effectiveness of a no-rinse cleansing foam plus moisturizer against a no-rinse pH-balanced liquid cleanser and barrier cream.[33] Two studies investigated the effectiveness of foam cleansers against soap and water and standard care.[19,31] Another study evaluated a one-step cleanser protectant lotion against wet wipes, dry wipes or wash cloths with facility wall dispensed soap[34]; and another investigated improvement of skin condition over time following use of 3M™ Cavilon Durable™ barrier cream (3M Australia, North Ryde, NSW) by comparing skin change over time using high-frequency ultrasound (HFU).[32] Description of studies including products, comparators and outcomes are provided in Appendix V.

Methodological quality

Methodological quality overall was poor, with few studies utilizing randomization, blinding or product allocation concealment. The large variation in comparators, outcomes, unclear or incomplete data and wide range of measurement tools made it difficult to conduct comparative effectiveness of interventions. Blinding of participants and consistency of interventions were subject to confounding owed to the nature and form of product application methods. An inability to control for reproducibility of intervention protocols due to clinicians' skill or experience, their practice variations and possible subjectivity of skin assessments were also likely confounders. Sample size ranged between 10 and 177 participants. Where individual randomization was not practical in studies using randomised control trial (RCT) study designs, cluster randomization by location, admission date and clinical settings was used.[5,19] One study commenced individual randomization but changed to randomization by setting within the care facility due to location of patients in different areas at different times of the day, for example bedroom at night and lounge or dining room during the day.[31] Data collected prior to change in randomization method may have likely contributed to confounding as no details of a washout period between products was reported.[31] Reporting of natural attrition and withdrawals and the inclusion of data on these participants in statistical analysis up to withdrawal were not always clearly discussed[31] (see Table 1).

Table 1: Assessment of methodological quality

Citation	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Experimental studies										
de Baatenburg and Admiraal[28]	Y	U	Y	U	N	Y	U	Y	Y	Y
Beeckman *et al.*[15]	Y	N	U	U	N	Y	N	Y	Y	Y
*Bohnenkamp *et al.*[29]										
Brunner *et al.*[30]	Y	N	N	Y	N	Y	N	Y	U	Y
Cooper and Gray[31]	U	N	N	U	Y	Y	Y	Y	Y	Y
Driver[7]										
Lewis-Byers and Thayer[33]	N	N	N	U	N	U	U	Y	Y	N
Park and Kim[19]	N	N	N	N	Y	Y	Y	Y	Y	Y
%	50	0	17	17	33	67	33	100	83	83
Citation	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	
Descriptive studies										
Kerr *et al.*[32]	N	Y	Y	Y	NA	Y	NA	Y	Y	
Warshaw *et al.*[34]	N	Y	Y	Y	NA	N	Y	Y	Y	
%	0	100	100	100	0	50	50	100	100	

*Unable to assess methodological quality as study presented as a poster.

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Findings of the review

Findings are narratively presented by types of interventions. Meta-analysis of the extracted results was not possible due to the heterogeneity of the measurement tools, outcomes, time points and interventions (Table 2).

3-in-1 wash cloths

Four studies (one quasi-experimental, one RCT and two pre-post studies) evaluated disposable perineal wash cloths impregnated with a cleanser, moisturizer and barrier.[7,15,29,30] The wash cloths were tested against pH neutral soap and water,[15] a two-step

Table 2: Skin assessment tools used in studies reviewed

Study	Skin assessment tool used
de Baatenburg and Admiraal (2004) Zinc Oxide oil vs Cavilon NSBF (No Sting Barrier Film)[28]	Skin condition assessment scale: measured area and severity of both redness and denudation on a 4-point Likert scale 0–3
Beeckman *et al.* (2011) Comfort Shield 3-in-1 wash cloths vs pH neutral soap and water[15]	IAD Skin Condition Assessment Tool (Kennedy Lutz 1996), Measured area redness in cm^2 degree of redness and erosion on a 4-point Likert scale 0–3
Bohnenkamp *et al.* (2013) All-in-1 washcloths vs standard care (no details provided)[32]	Self-developed using a computerized tablet program
Brunner *et al.* (2012) 3-in-1 wipe/wash cloths vs two-step cleanser + barrier film spray (NSBF)[30]	Self-developed skin classification tool
Cooper and Gray (2001) Clinisan foam cleanser with dimethicone vs soap and water[31]	Stirling Pressure Sore Severity Scale (Reid and Morrison 1994)
Driver (2007) 3-in-1 wash cloths with dimethicone vs no rinse cleanser + calmoseptine (zinc oxide)[7]	Self-developed skin classification tool
Kerr *et al.* (2014) Cutaneous hydration of skin with IAD treated with Cavilon Durable barrier cream (3M) application vs healthy adjacent skin[32]	IAD Skin Condition and Assessment Tool (Kennedy and Lutz 1996)
Lewis-Byers and Thayer (2002) No rinse skin liquid cleanser + barrier cream vs standard care (based on soap and water + moisturizing lotion)[33]	5-point Likert scale from 0 to 4 for erosion or ulceration
Park and Kim (2014) No rinse foam skin cleanser (Elta Cleansing Foam) vs "standard care" + soap and water[19]	IAD and its Severity Instrument (IADS) (Borchert 2010)
Warshaw *et al.* (2002) Perineal no-rinse cleanser protectant lotion (one step) vs Standard care: Wet wipes, dry wipes or wash cloths with soap and water and barrier cream (two step)[34]	Perineal Assessment Tool (PAT) (Nix 2002) and Erythema and Pain Grading Scale

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cleanser and barrier film spray,[30] a no-rinse cleanser and Calmoseptine® (zinc cream)[7] and a standard care regimen (no details provided)[29] (Table 3). There was a total of 535 participants across all studies with majority female participants in Beeckman *et al.*[15] as opposed to Brunner *et al.*[30] where there were more males and no demographic data on gender provided in the Driver *et al.* or Bohnenkamp *et al.* studies. Mean age of participants at entry across the studies, where these data were provided, was between 66.4 and 86.3 years. One study was conducted in four nursing homes in Belgium, and the others were set in the United States of America.[15,30] Of these three studies, two were conducted in intensive care unit (ICU) and one set in acute care that included ICU.

All four studies measured IAD prevalence or incidence as an outcome.[7,15,29,30] At day 120, the group treated with the Sage Comfort Shield® 3-in-1 wash cloths had six (8.1%) participants with IAD and the soap and water group 18 (27.1%).[15] The prevalence of IAD in the Bohnenkamp *et al.*[29] study was nine (81%) and one (9%) in the control/standard care group and experimental/wash cloth groups, respectively. Brunner *et al.*[30] found 27.3% of participants in the wipes/washcloth developed IAD as opposed to 22.6% of participants treated with the two-step cleanser + barrier film spray. The forth study by Driver[7] yielded a 19% IAD rate in the group treated with disposable perineal wash cloths saturated with a cleanser, moisturizer and barrier, and a 50% IAD rate in the group treated with a wash cloth and ZnO cream. These findings reveal that use of one-step wash cloths with built-in cleanser, moisturizer and barrier had a lower IAD prevalence and incidence rate when compared to soap and water or two-step cleanser and barrier film or zinc oxide cream.

The mean IAD severity score (using the IAD Skin Condition Assessment Tool[35]) was also higher in the soap and water group (6.4/10) and lower (3.9/10) in the group treated with Sage Comfort Shield® 3-in-1 wash cloths.[15] More participants in the 3-in-1 wipes group developed moderate skin breakdown ($n = 4$, 12.1%), and the time to skin breakdown was 91.1 hours. A significant intervention effect on IAD

Table 3: Results of 3-in-1 wash cloths versus other interventions

Intervention	Outcomes		Results *n* (%)	
			Exp. ($n = 73$) washcloths	Cont. ($n = 68$) Soap + water
Comfort Shield 3-in-1 wash cloths vs pH neutral soap and water[15]	IAD prevalence	Baseline	16 (22.3)	15 (22.8)
		Day 120	6 (8.1)	18 (27.1)
	IAD Severity Score	Baseline	6.9/10	7.3/10
		Day 120	3.9/10	6.4/10
One-step wash cloths vs standard care[29]	IAD prevalence		Exp. wash cloths Post intervention ($n = 11$)	Cont. standard care Pre-intervention ($n = 11$)
			1 (9)	9 (81)
			Exp. ($n = 33$) Washcloths/wipes	Cont. ($n = 31$) Cleanser + barrier film spray
3-in-1 wipe/wash cloths vs two-step cleanser + barrier film spray (No Sting Barrier Film)[30]	Intact skin		24 (72.7)	24 (77.4)
	Mild skin breakdown		5 (15.2)	6 (19.4)
	Moderate skin breakdown		4 (12.1)	1 (3.2)
	Severe skin breakdown		0 (0)	0 (0)
	Time to skin breakdown		91.1 h	213.3 h
Comfort shield (3M) 3-in-1 wash cloths vs No rinse cleanser + calmoseptine (zinc oxide)[7]	IAD prevalence		Phase 2 Exp. ($n = 177$) Washcloths	Phase 1 Cont. ($n = 131$) No rinse + calmoseptine
	No skin breakdown		13 (81)	8 (50)
	Skin breakdown		3 (19)	8 (50)

Cont., control; Exp., experimental; IAD, incontinence-associated dermatitis.

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prevalence was reported between the experimental group (3-in-1 wipes) 8.1% versus control group (soap and water) 27.1% ($F = 3.1$, $P = 0.003$).[15] However, a nonsignificant effect on IAD severity[35] was reported (experimental: 3.8/10 versus control: 6.9/10, $F = 0.8$, $P = 0.06$).[15] Brunner *et al.*[30] reported one incident of skin breakdown (3.2%) and 213.3 h to IAD occurrence in the cleanser + barbarrier film spray (NSBF) group. Average time to skin breakdown was significantly longer in NSBF group ($n = 6$; 213.3 h) versus 3-in-1 wipes group ($n = 6$; 91.1 h) (F [1, 11] = 5.27, $P = 0.045$).[30] Findings from these two studies suggest that 3-in-1 wash cloths might have been more effective in preventing IAD incidence and severity than soap and water but might not have been as effective as an NSBF in delaying skin breakdown.[7,15,29,30]

Foam cleansers

Two studies (one RCT and one quasi-experimental study) with a total of 163 participants tested the use of foam cleansers in comparison to soap and water and standard care.[19,31] The majority of participants were female across both studies.[19,31] The median age for both studies at entry to trial was 79 and 85 years, respectively, in the experimental and control groups in Cooper and Gray[31] and 66.2 and 69.7 years in Park and Kim.[19] One study was set in aged care in the United Kingdom, and the second in medical, cardiac, thoracic and neuro ICUs.[19,31]

The groups treated with the foam cleansers had a larger percentage of participants who maintained skin integrity: 80.4% compared to 37% in the soap and water group in Cooper and Gray[31] Of these two studies, Cooper and Gray[31] also measured skin erosion, skin ulceration or skin denudation specific to IAD, and Park and Kim[19] measured pressure injury (PI) development creating difficulty in conducting comparative analysis. In Cooper *et al.*, the proportion of participants who maintained skin integrity was 66% in the foam cleanser and 37% in the soap and water groups, respectively. A moderately higher number of participants (19.6%) had skin redness/erythema or rash in the control (soap and water) group as opposed to 12.2% in the experimental group. The difference between the number of participants who developed skin erosion/ulceration or denudation was small – two (7.4%): five (10.9%) in the control and three (7.3%) in the experimental group. Park *et al.* assessed IAD severity using the Identifying Incontinence-Associated Dermatitis and its Severity Instrument (IADS)[3] between the two groups.[19] The IADS instrument allocates a score ranging from 1 to 3 with a total of 16 for the anatomical areas with IAD, scores of 1 or 2 for skin color change and a score of 3 for skin loss. Based on these scores, IAD at its most extensive and severest will result in an IADS score of 21.[3] The IADS score in the experimental group was 5.19: standard deviation (SD) 3.41 and 14.3 SD 11.7 (P value <0.001) in the control group indicating that the foam cleanser group had less-severe IAD scores than the soap and water groups. Overall, foam cleansers appeared to be slightly more effective than soap and water in preventing or treating IAD or preventing skin breakdown that may lead to pressure ulcers. Results from these studies are detailed in Table 4.

Table 4: Results of foam cleansers versus soap and water

Intervention	Outcomes	Results *n* (%)	
Clinisan foam cleanser with dimethicone vs soap and water[31]		Exp. (n = 41) Clinisan	Cont. (n = 46) Soap and water
	Maintained skin integrity	27 (66)	17 (37)
	Skin redness/erythema/rash	5 (12.2)	9 (19.6)
	Skin erosion/skin ulceration/ skin denudation	3 (7.3)	5 (10.9)
	Skin deterioration	5 (12.2)	14 (30.4)
No rinse foam skin cleanser (Elta Cleansing Foam) vs "standard care" + soap and water[19]		Exp. (n = 38) Cleansing foam	Cont. (n = 38) Soap and water
	Pressure ulcer occurrence rate	5 (13.2)	19 (50)
	IAD severity score	M 5.19 SD 3.41	M 41.13 SD 11.7

Cont., control; Exp., experimental; IAD, incontinence-associated dermatitis; SD, standard deviation.

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No-Sting Barrier Film

Two studies with a total of 106 participants examined Cavilon™ NSBF as a skin protectant in comparison to ZnO oil (RCT) and Sage Comfort Shield® perineal wash cloths (quasi-experimental study) (Table 5).[28,30] It is unclear how participants in the de Baatenburg and Admiraal[28] study were allocated into the control and experimental groups with 19 participants in one and 20 in the other as this cannot be determined by the reported study methods. There were more females included in de Baatenburg and Admiraal,[28] and Brunner *et al.*[30] had more male participants. Outcomes measured by these studies differed, making it difficult to pool data and make comparisons on effectiveness. Participants across both studies were between 66.4 and 85.9 years.[28,30]

de Baatenburg and Admiraal[28] demonstrated that nine (47.4%) participants treated with ZnO developed skin redness/erythema or a rash as opposed to three (15%) in those treated with the Cavilon™ NSBF.[30] The second outcome measured area and severity of skin erosion, ulceration or denudation. Fewer participants in the Cavilon™ NSBF group ($n = 3$, 16.7%) experienced some degree of skin erosion, ulceration or denudation as opposed to those treated with ZnO, among whom eight (42.9%) developed skin erosion.[28] At end of trial, skin integrity deteriorated in two (8.3%) of those treated with NSBF as opposed to seven (35.7%) in those treated with ZnO.[28] Total skin damage scores improved with both treatments, but were significantly better with 3M™ Cavilon™ NSBF ($P = 0.04$).[28] In Brunner *et al.* (NSBF versus 3-in-1 wipes), as discussed previously in comparison to Beeckman *et al.*,[15] more participants in the 3-in-1 wipes group developed moderate skin breakdown ($n = 4$, 12.1%), and the time to skin breakdown was 91.1 hours as opposed to one incident of skin breakdown (3.2%) and 213.3 hours to skin breakdown in the cleanser + barrier film spray (3M™ Cavilon™ NSBF) group.[30] Average time to skin breakdown was significantly longer in 3M™ Cavilon™ NSBF group ($n = 6$; 213.3 h) versus 3-in-1 wipes group ($n = 6$; 91.1 h) ($F[1, 11] = 5.27$, $P = 0.045$).[30] These results suggest that 3M™ Cavilon™ NSBF was more effective in protecting or delaying skin breakdown than either ZnO oil and soap and water, and ZnO had better skin protectant properties than soap and water.[28,30]

Table 5: Results for No-Sting Barrier Film against other interventions

Intervention	Outcomes	Results *n* (%)	
		Exp. ($n = 19$) ZnO	Cont. ($n = 19$) NSBF
Zinc oxide oil vs Cavilon NSBF[28]	Skin redness/erythema/ rash	9 (47.4)	3 (15)
	Severity of skin erosion/skin ulceration/skin denudation/skin breakdown: day 1	8 (42.9)	3 (16.7)
	Severity of skin erosion/skin ulceration/skin denudation/skin breakdown: end of trial (day 14)	7 (35.7)	2 (8.3)
		Exp. ($n = 33$) Wipes	Cont. ($n = 31$) NSBF
Comfort shield perineal care wipes vs Cavilon NSBF[30]	Intact skin	24 (72.7)	24 (77.4)
	Mild skin breakdown	5 (15.2)	6 (19.4)
	Moderate skin breakdown	4 (12.1)	1 (3.2)
	Severe skin breakdown	0 (0)	0 (0)
	Time to skin breakdown in hours	91.1 h	213.3 h

Cont., control; Exp., experimental; NSBF, No Sting Barrier Film.

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No-rinse cleanser protectant lotion

Warshaw *et al.*, in a prospective quasi-experimental non-controlled study, and Lewis-Byers and Thayer, in a prospective descriptive two-group study, tested the efficacy of a no-rinse cleanser protectant lotion against standard care based on soap and water regimens (Table 6).[33,34] Both studies reported data on a combined total of 50 participants.[33,34] Warshaw *et al.*[34] used two separate sites for each intervention, and Lewis-Byers and Thayer[33] assigned odd- and even-numbered rooms to the two interventions. The Warshaw *et al.*[34] study had more male (73.7%) than female participants (26.3%), and the gender distribution in the Lewis-Byers and Thayer study was not reported.[33] The age range for all participants across both studies was 41 and 105 years.[33,34]

Warshaw *et al.* reported a decrease in Perineal Assessment Tool (PAT)[10] scores between day one and day seven from 2.3 to 0.6 in the experimental group with the baseline score reported at 5.9. The PAT measures risk of IAD on a scale of 1–3 for four items including intensity and duration with higher scores indicating more severe IAD. Erythema scores after seven days were lower: mean 0.6 and SD 0.8; however, data for PAT scores[10] in the control group were not provided.[34] Lewis-Byers and Thayer[33] reported that 13 (72.3%) participants in the experimental group maintained skin integrity, two (11%) with worsened skin condition and three (16.7%) with improved skin condition. In the control group, nine (69.2%) maintained skin integrity, three (23%) had worsened and one (7.7%) had improved.[33] Data at baseline and completion for the soap and water groups reported baseline mean skin condition scores as 0.54; SD 0.88 and at completion mean 0.77; SD 1.30. Data for the no-rinse cleanser protectant group at baseline was mean 0.56; SD 1.15 and at completion mean 0.28; SD 0.83.[33] These findings by Lewis-Byers and Thayer[33] suggest that the no-rinse cleanser lotion was marginally more effective at protecting the skin from IAD than soap and water skin care regimens, with only a 3.1% difference between the two groups maintaining skin integrity. In relation to treating IAD, Lewis-Byers and Thayer[33] reported a difference of 9% with improved skin condition favoring the no-rinse skin liquid cleanser over soap and water.

3M™ Cavilon™ durable barrier cream

In a prospective cohort descriptive study of 10 participants over 14 days, Kerr *et al.*[32] measured low echogenic pixel (LEP) imaging of affected skin and compared with the LEP of normal skin adjacent

Table 6: No rinse cleanser protectant lotion versus soap and water + standard care

Intervention	Outcomes	PAT score results (total PAT score = 12)	
Perineal no-rinse cleanser protectant lotion (one step) vs standard care: wet wipes, dry wipes, or wash cloths with soap and water and barrier cream (two step)[34]	PAT score Baseline PAT score 5.9 (12)	Exp. no rinse cleanser PAT scores	
		Day 1 = 2.3 (12)	
		Day 7 = 0.6 (12)	
		Cont. standard care No reported data	
No-rinse skin liquid cleanser + barrier cream vs standard care (based on soap and water + moisturizing lotion)[33]	Outcomes	Exp. (n = 18) No-rinse skin liquid cleanser	Cont. (n = 13) soap and water
	Maintained skin integrity	13 (72.3)	9 (69.2)
	Improvement in skin condition	3 (16.7)	1 (7.7)
	Worsened skin condition	2 (11%)	3 (23%)
	Skin redness/erythema at baseline	1 (5.6)	4 (30.8)

Cont., control; Exp., experimental; PAT, Perineal Assessment Tool.

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Table 7: Cavilon durable barrier cream

Intervention	Outcomes	Results of LEP M (SD)	
Cutaneous hydration of skin with IAD treated with Cavilon Durable barrier cream (3M) application vs healthy adjacent skin[32]	Decreased cutaneous hydration in skin with IAD as measured by a change in LEP over time to demonstrate improved skin condition and resolution of IAD using high-frequency ultrasound	LEP of normal skin using high-frequency ultrasound	0.123 (0.023)
		Visit 1	0.345 (0.076)
		Visit 2	0.255 (0.074)
		Visit 3	0.137 (0.037)

IAD, incontinence-associated dermatitis; LEP, low echogenic pixel; M (SD), mean (standard deviation).

to the affected area using HFU (Table 7). The age of participants ranged between 80 and 100 years with 60% female participants.[32]

Results showed the mean LEP measure of normal skin for the participants to be 0.123.[32] At the first visit, the mean LEP baseline measure of affected skin was 0.345.[32] Following treatment with 3M™ Cavilon™ durable barrier cream, the mean LEP measures dropped to 0.255 at day seven and 0.137 at end of trial on day 14.[32] The mean LEP at end of trial dropped to a level close to that of normal skin at baseline with a difference of 0.14 between the two readings.[32] These findings suggest that 3M™ Cavilon™ durable barrier cream was effective in treating and resolving IAD.[32] High-frequency ultrasound has been used to in dermatological practice for histological evaluation of skin lesions and tissue injury and can be used to measure hydration of skin with LEP, which is increased in dermal inflammation. It has however not been extensively used in clinical trials or practice to diagnose IAD.[36,37] Given the sample size for this study was very small, the findings may not be generalizable based on the fact that HFU and LEP have not been established as reliable measures to detect and diagnose IAD.

Discussion

At the time of conducting this systematic review, there were a limited number of clinical trials that compared products for efficacy in preventing and treating IAD. The study methodologies varied in the form of RCTs, quasi-experimental, non-controlled and prospective cohort descriptive studies.[6,15,19,28-34] A total of 804 participants were included across all 10 studies with the participant numbers ranging between 10[32] and 308[6] in individual studies. This is a relatively small number considering the fact that IAD prevalence rates are reported at 50–66% in aged care and community care residents experiencing incontinence.[2,3,38,39]

It is difficult to fathom the reasons for this lack of good-quality published evidence for such widely used products aiming to treat a common problem in such a large population. This lack of high-quality experimental studies is reiterated consistently in the literature.[40] A recent document published from the proceedings of a global IAD expert panel in 2015 identifies the comparison of effectiveness of different skin care products and skin care regimens as an area for future research on IAD.[20] Possible explanations for this lack of evidence are difficulty in double blinding of participants and care providers due to the presentation and packaging of products, difficulties with controlling individual clinicians' practice, controlling confounders, attitudes and practices that may affect results in clinical studies.

Furthermore, to determine whether an exposure is directly responsible for the outcome or to test efficacy of products that demonstrate a true cause and effect that is not affected by confounders may be difficult due to variances/limitations in study populations and nursing practice dissimilarities.[15,30,33] Confounders such as variances in the length of exposure to fecal matter following incontinent episodes, amount of friction used to clean skin and remove fecal matter, type of linen and disposable containment pads used, amount of skin protectant products used and variances in individual nurses' preferences and practices regarding skin care have the ability to affect results and were acknowledged as a limitation in some studies and not accounted for in others.[30,33] To overcome this, adequate sample

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sizes and randomization should enable researchers to account for these factors and establish a stronger estimate of a causal relationship between skin status and product use.[41]

Another possible reason may be that IAD is predominately a problem of the aged population that may be under-represented or concealed due to the negative connotations associated with incontinence. The lack of independent funding or the highly competitive nature of grant allocations where IAD research may be deemed less important to other medical conditions may be another noteworthy reason. Some studies in this review were funded by pharmacological companies with interests in skin products; however, this to a certain degree decreases confidence regarding objectivity and introduces conflict of interest.[15,28,34]

A consistent recurrent difficulty noted and acknowledged by studies in this review is the lack of a standardized skin assessment tool to diagnose and measure IAD. The studies included in this review used a wide variety of ways to measure IAD. Table 2 provides a list of the skin assessment tools used in the studies. Seven different skin assessment tools were used in the eight studies, with the IAD Skin Condition and Assessment Tool[35] used twice. Information on validity testing and reliability of skin assessment tools were limited. In addition to the lack of a standardized IAD measurement tool, the frequent misdiagnosis of IAD as pressure injuries and vice versa questions the reliability of IAD data provided in the studies as no distinction between the two conditions were addressed in any of the studies reviewed. Surprisingly, Cooper and Gray[31] used the Stirling Pressure Sore Severity Scale. This lack of consistent standardized IAD skin assessment reduces objectivity and opens IAD diagnosis to interpretation or misdiagnosis.

There is a general overall consensus that skin damage from exposure to urine and or fecal matter can be reduced by the implementation of structured skin care regimens that both cleanse and protect skin at risk.[20] In a literature review, Beeckman *et al.*[5] reiterated that structured skin care regimens that incorporate gentle cleansing and the use of skin protectants were shown to reduce the incidence of IAD. In the studies reviewed, soap and water regimens have shown to be least effective in protecting the skin against IAD.[15,19,31,33,34] No-rinse wipes or cleansers with built-in skin protectants were shown to have some benefit with a positive cost and time saving effect; however, statistical evidence on efficacy on skin protection would be more beneficial for clinicians and healthcare providers.[15,19,30,33,34] In view of the recognition of the positive effect of skin protectants on treating and preventing IAD, it is important to have some collaboration between researchers to encourage studies that exhibit some homogeneity that will allow statistical meta-analysis to be performed to demonstrate superior effectiveness, if any, between products.

In two independent systematic reviews of topical skin care in aged care (one by Hodgkinson *et al.* and the other by Beeckman *et al.*) focusing on IAD as risk factors for PI development, skin care was identified as a ''major'' issue for nursing.[40,42] Park and Kim[19] demonstrated that participants with higher IAD scores or skin damage scores were at higher risk of developing PU. From the literature, it is clear that IAD as a complication of incontinence needs more focus with regard to prevention and treatment. Although research on the condition has been initiated, the quality of product effectiveness when reviewed by these two reviews was weak due to small sample sizes and poor data reporting. Due to similar issues discussed in this review, Hodgkinson *et al.*[40] were unable to make recommendations but reiterated that more research into effectiveness of topical skin care products was needed.

Conclusion

Current research and studies indicate that skin care regimens that include the use of a skin protectant product are beneficial in preventing and treating IAD; however, with current data available, there is no proven evidence that indicates superior outcomes of any one product. Regardless of heterogeneity of skin assessment tools, study outcomes and methodology, most skin care products trialed (with the exception of soap and water regimens) demonstrated some skin protection qualities. However, more conclusive information on product supremacy is vital. This type of information can be valuable to both clinicians and healthcare organizations in assisting with the purchase and use of skin care protectants with both financial and clinical benefits. Hence, more RCTs on product effectiveness on prevention and treatment of IAD are highly recommended.

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Implications for practice

There is insufficient evidence to recommend a single skin care product for treating or preventing IAD in any care setting as the current evidence lacks rigor and consistency. Outcome measures were not homogeneous or truly objective, and a standardized IAD skin assessment tool was deficient across all studies.

Structured skin care regimens and use of skin products do to a certain degree prevent and protect skin against IAD development. The IAD proceedings from an expert panel indicate that dimethicone-containing products, acrylate terpolymer products and zinc oxide-based or petrolatum-based products may be beneficial and recommend them in their ''Cleanse, Protect and Restore'' program as part of interventions for prevention and management based on the severity of IAD.[20] However, the wide variety of products available and lack of effectiveness data create confusion and have potential cost implications if clinicians are not provided with conclusive product performance power data.

Implications for research

The use of consistent terminology for IAD as opposed to the wide variety of terms used to label the condition will assist research and improve education of healthcare providers. The testing, trialling and wide use of an industry-approved IAD skin assessment tool is also recommended to provide a standardized tool that can be used across all settings and by researchers for future studies. This will ensure the correct diagnosis and treatment for IAD and prevent it being misdiagnosed as a pressure injury. It is also important to consider and include studies in languages other than English, and an allocation of funding for publishing in multiple languages and translation should be included in research budgets and proposals. Considering the absence of high-quality true RCTs and the methodology limitations as evident in the studies included in this review, it is clear that more research comparing effectiveness of products with the measurement of common outcomes that can be statistically measured is required.

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Appendix I: Search strategy

PubMed (pubmed.gov)
Searched on August 24, 2015

Search	Query
#1	diaper dermatitis OR diaper erythema OR diaper rash napkin dermatitis OR napkin erythema OR napkin rash nappy dermatitis OR nappy erythema OR nappy rash perineal dermatitis OR perineal erythema OR perineal rash moist* sore* OR moist* ulcer* OR moist* damage OR moist* wound* Or moist* injur* OR moist* lesion* incontinen* dermatitis OR incontinen* sore* OR incontinen* ulcer* OR incontinen* damage OR incontinen* injur* Or incontinen* lesion* (MH "Skin Care+") skin care OR skin care regimen
#2	zinc oxide OR dimethicone OR petrolatum OR dermashield OR uniderm OR dermafilm OR cavilon OR aqueous cream OR sudocrem OR silver sulfadiazine OR lanolin topical treatment OR skin barrier OR skin protectant OR barrier cream OR emolient OR moisturiser OR ointment OR lotion OR cream OR barrier film OR moisture barrier OR humectant
#3	#1 AND #2
Limit to #(1980-present), English, adult participants >18 years of age, human research studies, limits#	

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Appendix II: MAStARI appraisal instrument

JBI Critical Appraisal Checklist for Randomised Control / Pseudo-randomised Trial

Reviewer _____ Date _____

Author _____ Year _____ Record Number _____

	Yes	No	Unclear	Not Applicable
1. Was the assignment to treatment groups truly random?	☐	☐	☐	☐
2. Were participants blinded to treatment allocation?	☐	☐	☐	☐
3. Was allocation to treatment groups concealed from the allocator?	☐	☐	☐	☐
4. Were the outcomes of people who withdrew described and included in the analysis?	☐	☐	☐	☐
5. Were those assessing outcomes blind to the treatment allocation?	☐	☐	☐	☐
6. Were the control and treatment groups comparable at entry?	☐	☐	☐	☐
7. Were groups treated identically other than for the named interventions	☐	☐	☐	☐
8. Were outcomes measured in the same way for all groups?	☐	☐	☐	☐
9. Were outcomes measured in a reliable way?	☐	☐	☐	☐
10. Was appropriate statistical analysis used?	☐	☐	☐	☐
	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info. ☐

Comments (Including reason for exclusion)

_____

_____

_____

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Findings	Illustration from Publication (page number)	Evidence		
		Unequivocal	Credible	Unsupported

Extraction of findings complete Yes ☐ No ☐

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Appendix III: MAStARI data extraction instrument

JBI Data Extraction Form for Experimental / Observational Studies

Reviewer _____ Date _____

Author _____ Year _____

Journal_____ Record Number_____

Study Method

RCT ☐ Quasi-RCT ☐ Longitudinal ☐

Retrospective ☐ Observational ☐ Other ☐

Participants

Setting

Population

Sample size

Group A _____ Group B _____

Interventions

Intervention A

Intervention B

Authors Conclusions:

Reviewers Conclusions:

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Study results

Dichotomous data

Outcome	Intervention ( ) number / total number	Intervention ( ) number / total number

Continuous data

Outcome	Intervention ( ) number / total number	Intervention ( ) number / total number

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Conclusions	Illustration from Publication (page number)	Evidence		
		Unequivocal	Credible	Unsupported

Include Yes ☐ No ☐

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Appendix IV: Excluded studies

Bliss Donna Z, Zehrer C, Savik K, Smith G, Hedblom. E. An economic evaluation of four skin damage prevention regimens in nursing home residents with incontinence.
Reason for exclusion: Focused on economic evaluation no clinical outcomes reported.

Buckley B, Dofitas R, Baltazar W, Quiambao P, Razor B. Topical zinc oxide based creams in a structured care regimen for the treatment of incontinence associated dermatitis in hospitalized adults band older children: a randomized controlled trial.
Reason for exclusion: Not study population: Children >12 years included.

Dofitas Rodney B. A randomized controlled clinical study comparing the efficacy and safety of Calmoseptine vs. Destin Maximum Strength Diaper Rash Paste in treatment of incontinence associated dermatitis in older children and adults.
Reason for exclusion: Not study population (included participants 12–17 years of age).

Eigsti, Janice E. Beds, baths, and bottoms: a quality improvement initiative to standardize use of beds, bathing techniques, and skin care in a general critical-care unit.
Reason for exclusion: Outcomes of interest not clearly defined.

Fletcher J. The use of skin barrier creams in patients with incontinence.
Reason for exclusion: Not research focused on one product with no statistical data.

Foureur N, Vanzo B, Meaum S and Senet P. Prospective etiological study of diaper dermatitis in the elderly.
Reason for exclusion: Not intervention of interest.

Lake L, Philbin S, Hargreaves T. Use of barrier-impregnated cloths to treat severe incontinence-associated dermatitis: a case study.
Reason for exclusion: Not study type of interest

Markham T, Kennedy F. Topical sucralfate for erosive irritant diaper dermatitis.
Reason for exclusion: Not study type of interest.

Motta Glenda J. An open-label clinical case series to demonstrate efficacy of the Kendall skin wellness system.
Reason for exclusion: Trial protocol, no results reported.

Ramirez Razor B, Buckley B, Quiambao P, Dofitas R, Baltazar W. Incontinence-associated dermatitis (IAD) study: Blinded assessment and treatment with zinc oxide-based ointment.
Reason for exclusion: Not population of interest (includes participants 12–17 years).

Rees J and Pagnamenta F. Reducing incontinence dermatitis.
Reason for exclusion: Not study type of interest.

Shelly S. consistency is the key to treating severe incontinence-associated dermatitis (IAD).
Reason for exclusion: Not study type of interest.

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Appendix V: List of study findings/conclusions/characteristics of included studies

Reference	Study design	Intervention	Setting	Sample size	Type of incontinence	Outcome measures	Follow-up duration	Skin assessment tool
de Baatenburg and Admiraal [28]	RCT	Zinc oxide oil vs Cavilon NSBF (no sting barrier film) (3M)	Netherlands in an acute care hospital	38	Urine and/or feces	Skin redness and denudation	14 day study with skin assessments in days 0, 1,2,7 and 14	Skin condition assessment scale: measured area and severity of both redness and denudation on a 4 point Likert scale 0–3
Beeckman *et. al*[15]	RCT	Comfort shield 3 in 1 wash cloths vs pH neutral soap and water	Belgium 11 Nursing homes wards Across 4 nursing homes	141	Urine and/or feces	IAD prevalence IAD severity	Study over 120 days	Skin Assessment Tool (SAT) (Kennedy Lutz 1996), Measured area redness in cm^2 degree of redness and erosion on a 4 point Likert scale 0–3
Bohnenkamp *et al.*[29]	Quasi Exp. pre-post study	All-in-1 washcloths vs standard care (no details provided)	USA 20 bed ICU	22	Not reported	IAD prevalence	30 days	Self-developed using a computerized tablet program
Brunner *et al.*[30]		3-in-1 wipe/wash cloths vs two-step cleanser + barrier film spray (NSBF) (3M)	USA ICU and acute care	64	Not reported	IAD prev. IAD severity Time to IAD development	Not reported	Self-developed skin classification tool
Cooper and Gray[31]	RCT	Clinisan foam cleanser with dimethicone vs soap and water	UK Aged care	87	Urine and/or feces	Erythema Broken skin	10 months	Self-developed skin classification tool
Driver[7]	Quasi Exp. Pre-post study	3-in-1 wash cloths with dimethicone vs No rinse cleanser + Calmoseptine (zinc oxide	USA 28 bed ICU	308	Fecal	Skin breakdown	≤14 days >14 days	Self-developed skin classification tool
Kerr *et al.*[32]	Prospective cohort descriptive	Cutaneous hydration of skin with IAD treated with Cavilon Durable barrier cream (3M) application vs healthy adjacent skin	UK Aged care	10	Urine and/or fecal	Skin excoriation using ultrasound	14 days	IAD Skin Condition and Assessment Tool (Kennedy and Lutz 1996)
Lewis-Byers and Thayer[33]	Randomized prospective descriptive study	No rinse skin liquid Cleanser + barrier cream vs Standard care (based on soap and water + moisturizing lotion)	UK Aged care	41	Urine and/or fecal	perineal skin integrity	3 weeks	5 point Likert scale from 0 to 4 for erosion or ulceration
Park and Kim[19]	Non randomized quasi-experimental cohort study	No rinse foam skin cleanser (Elta Cleansing Foam) vs "standard care" + soap and water	South Korea Across 5 ICUs	76	Fecal incontinence	Mean IADS score Erythema Rash Skin loss	7 days	IAD and its Severity Instrument (IADS) (Borchert 2010)
Warshaw *et. al.*[34]	Prospective Quasi-experimental non-controlled	Perineal no-rinse cleanser protectant lotion (one step) vs Standard care: wet wipes, dry wipes or wash cloths with soap and water and barrier cream (two step)	USA long-term acute care and aged care facility for veterans	19	Urine and fecal	Erythema score	6 days	Perineal Assessment Tool (PAT) score (NIX 2002)

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